Sector decomposition for an anisotropic phase

A stationary phase bound, sector by sector and uniformly in x, for $I(t,x) = \displaystyle\int_{\mathbb{R}^d} e^{i(x\cdot\xi-\psi(t,\xi))}\,|A_t\xi|^{-1}\chi(\xi)\,d\xi.$

Every symbol is defined on this sheet, and every step comes with its reason.

A THE PROBLEM

Fix exponents $\ell_1,\dots,\ell_d \ge 0$ and, for $s>0$, let $A_s = \operatorname{diag}(s^{\ell_1},\dots,s^{\ell_d})$. A wave moves with speed s^{ℓ_j} in direction j, so after time t it carries the phase

$$\psi(t,\xi) = \int_1^t |A_s\xi|\,ds.$$

For $t \ge 2$ and $x \in \mathbb{R}^d$ we study

$$I(t,x) = \int_{\mathbb{R}^d} e^{i(x\cdot\xi-\psi(t,\xi))}\,|A_t\xi|^{-1}\,\chi(\xi)\,d\xi,$$

where χ is smooth and supported in $\frac12 \le |\xi| \le 2$.
Goal. Bound $\sup_x |I(t,x)|$ for large t, one sector at a time.
Directions seen from sector k. For $j \ne k$ put

$$\kappa_j = 2\ell_j - \ell_k + 1.$$

Direction j is *good* if $\ell_j \ge \ell_k$ and *bad* if $\ell_j < \ell_k$. It is *oscillatory* if $\kappa_j > 0$ and *non-oscillatory* if $\kappa_j \le 0$.

- $\mathcal{O}$: oscillatory directions. Every good direction is in $\mathcal{O}$, since $\kappa_j \ge \ell_k + 1 > 0$.
- $\mathcal{T}$: non-oscillatory directions. They are all bad.
- $\mathcal{B} = \mathcal{O} \cap \{\text{bad}\}$: bad oscillatory directions.

B THE RESULT

Theorem (main estimate). There are t_0 and C, depending only on d and the ℓ_j, such that for $t \ge t_0$, all x and all $\xi_k \ne 0$,

$$\Big|\int_{\mathbb{R}^{d-1}} e^{i\lambda\Phi(\eta)}\,b(\xi_k,\eta)\,d\eta\Big| \le C\,\lambda^{-\frac{d-1}{2}} \prod_{j\in\mathcal{T}} t^{\kappa_j/2}\,|\xi_k|^{-1/2},$$

with $\lambda = t^{\ell_k+1}|\xi_k|$.

Here η, Φ and b are the variables, phase and amplitude of the inner integral of sector k (Step 1).
Reading. If $\mathcal{T}$ is empty, this is the full stationary phase rate $\lambda^{-(d-1)/2}$. Each factor from $\mathcal{T}$ is at most $|\xi_k|^{-1/2}$.
The outer ξ_k integral, where $|\xi_k|$ can be as small as $t^{\ell_{\min}-\ell_k}$, is not treated here.

C WHY IT IS DELICATE

Each direction has its own speed s^{ℓ_j}. The phase is not radial, so no single large parameter sits in front of it.

In a bad direction, near $\eta_j = 0$ the coordinate j dominates at early times. High derivatives of the phase then blow up as $t \to \infty$.

Each ∂_{η_j} on the cutoff costs $t^{\ell_k-\ell_j}$, which is huge for bad j. A direct stationary phase bound loses $t^{d(\ell_k-\ell_{\min})}$.

0 THE TOOL

A stationary phase bound that does not see the size of the domain.

Theorem 0 (Alazard, Burq, Zuily). Let $b \in C_0^\infty(\mathbb{R}^n)$ have support K, and let K_1 be the points at sup distance at most 1 from K. Let Φ be real and smooth near K_1, with

$$a_0 = \inf_{K_1} |\det \nabla^2\Phi| > 0$$

and $\nabla\Phi$ injective on K_1. Then for $\lambda \ge 1$,

$$\Big|\int e^{i\lambda\Phi}b\Big| \le \frac{C_n}{a_0}\,(1+M_{n+2})^{n/2}\,N_{n+1}\,\lambda^{-n/2},$$

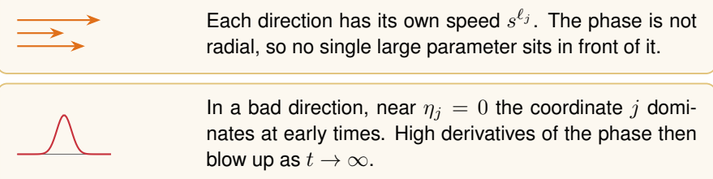
where $M_m = \sum_{2\le|\alpha|\le m} \sup_{K_1} |\partial^\alpha\Phi|$ and $N_m = \sum_{|\alpha|\le m} \sup_K |\partial^\alpha b|$.

Why it fits. No volume of K appears, so a large support costs nothing. Φ enters only through derivatives of order at least two, so the linear term that carries x is invisible.

T. Alazard, N. Burq, C. Zuily, *A stationary phase type estimate*, Proc. AMS 145 (2017), Theorem 3.

→ THE CHAIN OF VARIABLES

The proof uses three coordinate systems, in this order, each for one job.



The four substitutions, in order.

1. Chart $\eta_j = t^{\ell_j-\ell_k}\xi_j/\xi_k$	outer ξ_k integral, inner η integral over $\lvert\eta\rvert \le 2\sqrt{d}$
2. Time $s = \sigma t$ inside ψ	$\psi = \lambda\,G(t,\eta)$: one large parameter $\lambda = t^{\ell_k+1}\lvert\xi_k\rvert$
3. Shift $x \to A_t x$	x enters only through a term linear in η
4. Zoom $\eta = \lambda^{-1/2}\omega$, ξ_k fixed	decay $\lambda^{-\frac{d-1}{2}}$ in front; derivatives at the cell scale

Why η before ω. G has its t-uniform properties in η, and λ depends on ξ_k, so the zoom comes only once ξ_k is fixed.
Helpers, which change no integral: $u = D^{1/2}w$ puts the Hessian in Cauchy–Schwarz form (Step 2), and $y_j = \sigma^{\ell_j-\ell_k}\eta_j$ turns H into $(1+|y|^2)^{1/2}$ (Step 3).

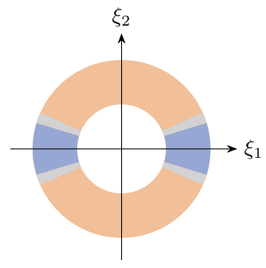
All together. Substitutions 1 and 4 give, for $j \ne k$,

$$\xi_j = \pm\, t^{-\kappa_j/2}\,|\xi_k|^{1/2}\,\omega_j.$$

A unit step in ω_j is a step of $t^{-\kappa_j/2}|\xi_k|^{1/2}$ in ξ_j, small compared with the scale 1 of χ exactly when $\kappa_j > 0$ (panel D).

1 STEP 1 · SECTORS

Sectors. Sector k is where $t^{\ell_k}|\xi_k|$ is the largest weight $t^{\ell_j}|\xi_j|$. There $t^{\ell_k}|\xi_k| \ge |A_t\xi|/\sqrt{d}$.
Smooth sectors. Take ρ_k on $\mathbb{S}^{d-1}$ with $\sum_k \rho_k = 1$ and $\rho_k(\theta) = 0$ unless $|\theta_k| \ge 1/(2\sqrt{d})$, and insert $\rho_k(A_t\xi/|A_t\xi|)$.



$d = 2$, $\ell_1 < \ell_2$.
blue: sector 1, coral: sector 2, gray: shared.

Chart. Compare each weight with the leading one:

$$\eta_j = \frac{t^{\ell_j}\xi_j}{t^{\ell_k}\xi_k}, \qquad \text{that is} \qquad \xi_j = t^{\ell_k-\ell_j}\xi_k\,\eta_j \quad (j \ne k).$$

Lemma 1 (normal form). $\psi(t,\xi) = \lambda\,G(t,\eta)$ with $\lambda = t^{\ell_k+1}|\xi_k|$ and

$$G(t,\eta) = \int_{1/t}^1 \sigma^{\ell_k} H\,d\sigma, \qquad H = \Big(1 + \sum_{j\ne k} \sigma^{2(\ell_j-\ell_k)}\eta_j^2\Big)^{1/2}.$$

After $x \to A_t x$, the sector integral is

$$I_k = t^{\,d\ell_k - \sum_j \ell_j - \ell_k} \int_{\mathbb{R}} e^{ix_k t^{\ell_k}\xi_k}\,|\xi_k|^{d-2} J(\xi_k)\,d\xi_k,$$

$$J = \int_{\mathbb{R}^{d-1}} e^{i\lambda\Phi(\eta)} b(\xi_k,\eta)\,d\eta, \quad \Phi = \frac{\operatorname{sgn}\xi_k}{t}\sum_{j\ne k} x_j\eta_j - G.$$

Why. $|A_s\xi| = s^{\ell_k}|\xi_k|\big(1+\sum_j (s/t)^{2(\ell_j-\ell_k)}\eta_j^2\big)^{1/2}$; then put $s = \sigma t$. The Jacobian is $d\xi = t^{\,d\ell_k - \sum_j \ell_j}|\xi_k|^{d-1}d\xi_k\,d\eta$.
Amplitude. $b = (1+|\eta|^2)^{-1/2}\tilde\chi\,\rho_k(\cdots)$, where $\tilde\chi$ is χ in the new variables. It is smooth and supported in $|\eta| \le 2\sqrt{d}$. On its support

$$\frac{t^{\ell_{\min}-\ell_k}}{2\sqrt{d}} \le |\xi_k| \le 2, \qquad \lambda \ge c\,t^{1+\ell_{\min}}.$$

For fixed ξ_k the inner integral has the single large parameter λ, and x enters only through a linear term.

2 STEP 2 · CURVATURE

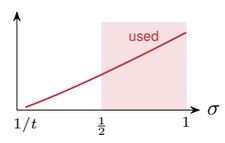
Every time adds curvature. With $D(\sigma) = \operatorname{diag}(\sigma^{2(\ell_j-\ell_k)})_{j\ne k}$,

$$-\nabla_\eta^2\Phi = \int_{1/t}^1 \sigma^{\ell_k} H^{-3}\big(H^2 D - (D\eta)(D\eta)^T\big)\,d\sigma.$$

Put $u = D^{1/2}w$ and $v = D^{1/2}\eta$, so $|v|^2 = H^2 - 1$. Cauchy–Schwarz gives $\langle v,u\rangle^2 \le (H^2-1)|u|^2$, hence

$$\int_{1/t}^1 \sigma^{\ell_k} H^{-3}|u|^2 d\sigma \le -\langle\nabla^2\Phi\,w,w\rangle \le \int_{1/t}^1 \sigma^{\ell_k} H^{-1}|u|^2 d\sigma.$$

Late times. On $\sigma \in [\frac12,1]$ every power of σ is comparable to 1, and $H \le C_R$ on $|\eta| \le R$. The integrand is nonnegative, so this piece alone gives a lower bound.



Lemma 2 (curvature). Φ is strictly concave in η for every t, x, ξ_k, and $-\nabla_\eta^2\Phi \ge c_R\,I$ on $|\eta| \le R$, with c_R independent of t.

Consequence. $\langle\nabla\Phi(\eta) - \nabla\Phi(\eta'), \eta-\eta'\rangle < 0$ for $\eta \ne \eta'$, so $\nabla\Phi$ is injective on all of $\mathbb{R}^{d-1}$. No localization near critical points is needed.

3 STEP 3 · DERIVATIVES

The problem. If $\ell_j < \ell_k$, early times magnify η_j by $\sigma^{\ell_j-\ell_k} \gg 1$. On the η_j axis,

$$\partial_{\eta_j}^{2N} G(t,0) = (2N)!\binom{1/2}{N}\int_{1/t}^1 \sigma^{\ell_k+2N(\ell_j-\ell_k)}d\sigma \to \infty$$

as $t \to \infty$, as soon as the exponent is below -1.
The cure. Measure derivatives in units of $|\eta_j|$: put $w_j = 1$ if j is good and $w_j = |\eta_j|$ if j is bad.

Lemma 3 (derivatives). Let $|\alpha| \ge 2$ with α supported in $\mathcal{O}$, and write $\alpha = \delta_i + \delta_j + \alpha'$ with $i,j \in \mathcal{O}$ and $\alpha' \ge 0$. Then for $t \ge 2$,

$$|\partial_\eta^\alpha\Phi| = |\partial_\eta^\alpha G| \le C_\alpha \prod_{m\ne k} w_m^{-\alpha'_m}.$$

Why. With $\mu_m = \sigma^{\ell_m-\ell_k}$ and $y_m = \mu_m\eta_m$, $H = (1+|y|^2)^{1/2}$ behaves like a function homogeneous of degree one, so $|\partial_\eta^\alpha H| \le C\mu^\alpha H^{1-|\alpha|}$. Each factor $\mu_m/H \le w_m^{-1}$: for good m, $\mu_m \le 1 \le H$, and for bad m, $H \ge \mu_m|\eta_m|$. The two factors for i,j are paid by

$$\int_0^1 \sigma^{\ell_k}\mu_i\mu_j\,d\sigma \le \tfrac12\int_0^1 \big(\sigma^{2\ell_i-\ell_k} + \sigma^{2\ell_j-\ell_k}\big)d\sigma < \infty,$$

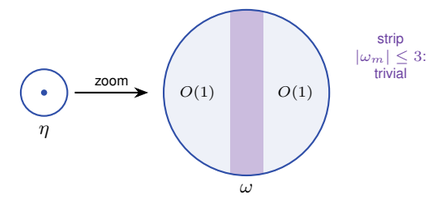
which is finite exactly because $\kappa_i, \kappa_j > 0$.
Two derivatives in oscillatory directions are free. Each further derivative in a bad direction costs $1/|\eta_m|$. Non-oscillatory directions are never differentiated.

4 STEP 4 · ZOOM

Zoom. Fix ξ_k and put $\eta = \lambda^{-1/2}\omega$, the stationary phase scale:

$$J = \lambda^{-\frac{d-1}{2}}\int e^{i\tilde\Phi(\omega)}\,\tilde b(\omega)\,d\omega, \qquad \tilde\Phi(\omega) = \lambda\,\Phi(\lambda^{-1/2}\omega).$$

Scaling. $\partial_\omega^\alpha\tilde\Phi = \lambda^{1-|\alpha|/2}\,\partial_\eta^\alpha\Phi$. The Hessian is unchanged and higher derivatives shrink. All the decay sits in front, and the ω integral only has to be bounded: Theorem 0 with $\lambda = 1$.



What one ω_j derivative costs.

falls on	cost	bounded when
cutoff $\tilde\chi$	$t^{-\kappa_j/2}\lvert\xi_k\rvert^{1/2}$	$j \in \mathcal{O}$
ρ_k, $(1+\lvert\eta\rvert^2)^{-1/2}$	$\lambda^{-1/2}$	always
phase, bad m (Lemma 3)	$\lvert\omega_m\rvert^{-1}$	$\lvert\omega_m\rvert \ge 1$

In ξ: a unit cell in ω_j has width $t^{-\kappa_j/2}|\xi_k|^{1/2}$ in ξ_j, while χ varies on scale 1.

5 STEP 5 · STRIPS

Strip cutoffs. Take $\vartheta \in C_0^\infty(\mathbb{R})$ with $\vartheta = 1$ on $[-2,2]$ and $\vartheta = 0$ outside $[-3,3]$. For each $m \in \mathcal{B}$ write $1 = \vartheta(\omega_m) + (1-\vartheta(\omega_m))$ and expand. This gives $2^{|\mathcal{B}|}$ pieces J_S, where $S \subset \mathcal{B}$ lists the directions placed in strips.
Active and frozen. In J_S the active variables are $A = \mathcal{O} \setminus S$. The variables in $S \cup \mathcal{T}$ are frozen and integrated last.
1. Theorem 0 in ω_A, with $\lambda = 1$.

$a_0 \ge c^{\lvert A\rvert}$	Lemma 2 on $\lvert\eta\rvert \le 3\sqrt{d}$; a principal submatrix of a matrix $\ge cI$ is $\ge cI$
injective	Lemma 2: $\tilde\Phi$ is strictly concave in ω_A
$\tilde M \le C$	Lemma 3: $\lambda^{1/2}w_m = \lambda^{1/2} \ge 1$ if m good, $= \lvert\omega_m\rvert \ge 1$ off the strips
$\tilde N \le C$	$t^{-\kappa_j/2}\lvert\xi_k\rvert^{1/2} \le \sqrt{2}$ for $j \in \mathcal{O}$

So the ω_A integral is at most C, uniformly in x and in the frozen variables.
2. Trivial integration in the frozen variables.

- $m \in S$: $|\omega_m| \le 3$, a factor 6.
- $j \in \mathcal{T}$: $|\xi_j| \le 2$ gives $|\omega_j| \le W_j = 2\,t^{\kappa_j/2}|\xi_k|^{-1/2}$, a factor $2W_j$.

A strip has width $\lambda^{-1/2}$ in η: exactly the gain oscillation would give.

6 STEP 6 · CONCLUSION

Sum the pieces. Each piece satisfies

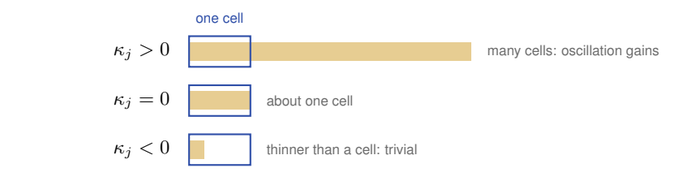
$$|J_S| \le C\,\lambda^{-\frac{d-1}{2}}\,6^{|S|}\prod_{j\in\mathcal{T}} 2W_j,$$

and summing over the $2^{|\mathcal{B}|}$ subsets S gives the main estimate in box B.
Uniform in x. x enters Φ only linearly, and none of a_0, $\tilde M$, $\tilde N$ sees a linear term.
Direct versus zoomed.

	Theorem 0 in η	Theorem 0 in ω
amplitude N	$\le C\,t^{d(\ell_k-\ell_{\min})}$	$\le C$
phase M	not uniform in t	$\le C$ off strips

D WHERE THE GAIN COMES FROM

In direction j, the support of the amplitude has width $W_j \approx t^{\kappa_j/2}|\xi_k|^{-1/2}$ measured in stationary phase cells.



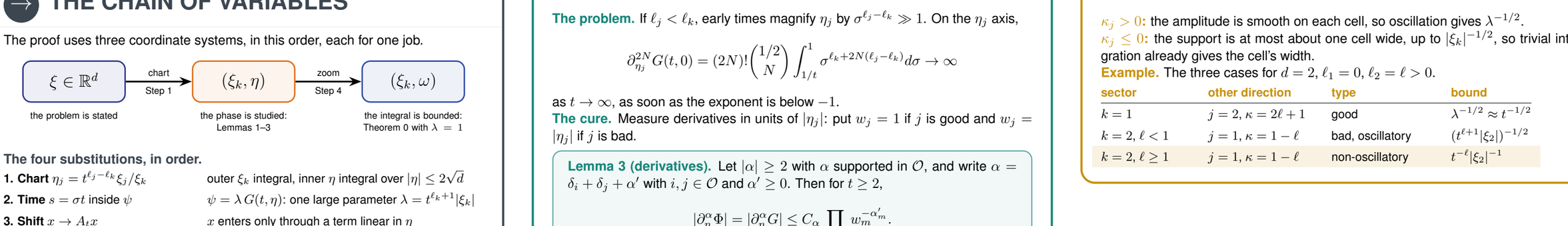
$\kappa_j > 0$: the amplitude is smooth on each cell, so oscillation gives $\lambda^{-1/2}$.
$\kappa_j \le 0$: the support is at most about one cell wide, up to $|\xi_k|^{-1/2}$, so trivial integration already gives the cell's width.
Example. The three cases for $d = 2$, $\ell_1 = 0$, $\ell_2 = \ell > 0$.

sector	other direction	type	bound
$k = 1$	$j = 2$, $\kappa = 2\ell + 1$	good	$\lambda^{-1/2} \approx t^{-1/2}$
$k = 2$, $\ell < 1$	$j = 1$, $\kappa = 1 - \ell$	bad, oscillatory	$(t^{\ell+1}\lvert\xi_2\rvert)^{-1/2}$
$k = 2$, $\ell \ge 1$	$j = 1$, $\kappa = 1 - \ell$	non-oscillatory	$t^{-\ell}\lvert\xi_2\rvert^{-1}$

i NOTATION AT A GLANCE

k, η_j, λ	sector, chart ratios, large parameter $t^{\ell_k+1}\lvert\xi_k\rvert$	1
G, H, Φ, b	normal form, its integrand, inner phase, amplitude	1
κ_j	$2\ell_j - \ell_k + 1$; its sign decides the type of direction j	A
$\mathcal{O}$, $\mathcal{T}$, $\mathcal{B}$	oscillatory, non-oscillatory, bad oscillatory	A
w_j	derivative weight: 1 if good, $\lvert\eta_j\rvert$ if bad	3
ω, $\tilde\Phi$, $\tilde b$	zoomed variable, phase, amplitude	4
ϑ, S, A, W_j	strip cutoff, strip set, active set, width in $\mathcal{T}$	5

★ THE WHOLE PROOF IN SIX MOVES



After zooming to the stationary phase scale $\lambda^{-1/2}$, the condition $2\ell_j - \ell_k + 1 > 0$ ***keeps the amplitude smooth on each cell.***
Where it fails, ***the support is already thinner than one cell****, so that direction needs no oscillation.*